

Astro HW 12

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12.1 Eddington luminosity and massive stars

a)

Given ...

$$L_{\text{Edd}} \simeq 3.3 \times 10^4 \left(\frac{M}{M_{\odot}} \right) L_{\odot}$$

$$L \simeq L_{\odot} \left(\frac{M}{M_{\odot}} \right)^3$$

$$L = L_{\text{Edd}}, \quad \text{when disruption occurs}$$

We can derive ...

$$\Rightarrow L_{\odot} \left(\frac{M}{M_{\odot}} \right)^3 = 3.3 \times 10^4 \left(\frac{M}{M_{\odot}} \right) L_{\odot}$$

$$\Rightarrow \left(\frac{M}{M_{\odot}} \right)^2 = 3.3 \times 10^4$$

$$\Rightarrow M_{\text{max}} \approx \sqrt{3.3 \times 10^4} M_{\odot} \approx 1.8 \times 10^2 M_{\odot}$$

b)

This calculated value of $M_{\text{lim}} \approx 1.8 \times 10^2 M_{\odot}$ is much lower than a considerable amount of observed systems. For example, the current most massive star has a proposed mass of $346 \pm 42 M_{\odot}$ [1].

12.2 Accretion disk temperatures

a)

Given ...

$$T \approx \left[\frac{GM}{2\sigma_{\text{SB}} r^2} \Sigma v \right]^{1/4}$$

$$\dot{M} = 2\pi r \Sigma(r) v(r)$$

We find ...

$$\dot{M} = 2\pi r \Sigma(r) v(r)$$

$$\Sigma(r) v = \frac{\dot{M}}{r 2\pi}$$

$$T \approx \left[\frac{GM}{2\sigma_{SB}r^2} \frac{\dot{M}}{r2\pi} \right]^{1/4}$$

$$T \approx \left[\frac{GMM\dot{M}}{4\pi\sigma_{SB}r^2} \right]^{1/4}$$

b)

Given ...

$$T \approx \left[\frac{GMM\dot{M}}{4\pi\sigma_{SB}r^2} \right]^{1/4}$$

$$L = \eta\dot{M}c^2$$

$$L = E_L, \quad \eta \sim .1, \quad r = \frac{6GM}{c^2}$$

c)

Sources

- [1] Z. Keszthelyi et al. “Evolutionary models for the very massive stars in the R136 cluster of 30 Doradus in the Large Magellanic Cloud”. In: *Astronomy and Astrophysics* 700 (Aug. 2025), A186. ISSN: 1432-0746. DOI: 10.1051/0004-6361/202554758. URL: <http://dx.doi.org/10.1051/0004-6361/202554758>.